Blockchain: Smart Contracts
Lecture 7

Ethereum: mechanics

Consensus (SMR): quick summary

Goals: **Safety** ($LOG_t^i \preccurlyeq LOG_s^j$ or $LOG_s^j \preccurlyeq LOG_t^i$) and **Liveness** (no censorship)

Network models: synchronous vs. asynchronous

Settings:

- Open participation: need sybil resistance (PoW or PoS)
- We saw two types of consensus protocols:
  - **Nakamoto-style consensus**: longest (heaviest) chain fork choice rule
    - dynamic availability, but no finality (no safety when asynchronous)
  - **PBFT-style** (block finalized when ≥t votes): finality and accountable safety
- Ethereum proof-of-stake: a finality chain that is a prefix of an available chain

New topic: limitations of Bitcoin

Recall: UTXO contains (hash of) ScriptPK

- simple script: indicates conditions when UTXO can be spent

Limitations:

- Difficult to maintain state in multi-stage contracts
- Difficult to enforce global rules on assets

A simple example: rate limiting. My wallet manages 100 UTXOs.

- Desired policy: can only transfer 2BTC per day out of my wallet

An example: DNS

Domain name system on the blockchain: [google.com → IP addr]

Need support for three operations:

- **Name.new**(OwnerAddr, DomainName): intent to register
- **Name.update**(DomainName, newVal, newOwner, OwnerSig)
- **Name.lookup**(DomainName)

Note: also need to ensure no front-running on **Name.new**()

A broken implementation

Name.new() and Name.upate() create a UTXO with ScriptPK:

DUP HASH256 <OwnerAddr> **EQVERIFY CHECKSIG VERIFY** <DNS> <DomainName> <IPaddr> <1>

only owner can "spend" this UTXO to update domain data

Contract: (should be enforced by miners)

if domain google.com is registered,
no one else can register that domain

verify
sig is valid

ensure top
of stack is 1

Problem: this contract cannot be enforced using Bitcoin script

What to do?

NameCoin: a fork of Bitcoin that implements this contract

(see also the Ethereum Name Service -- ENS)

Can we build a blockchain that natively supports generic contracts like this?

$\Rightarrow$ Ethereum

Ethereum: enables a world of applications

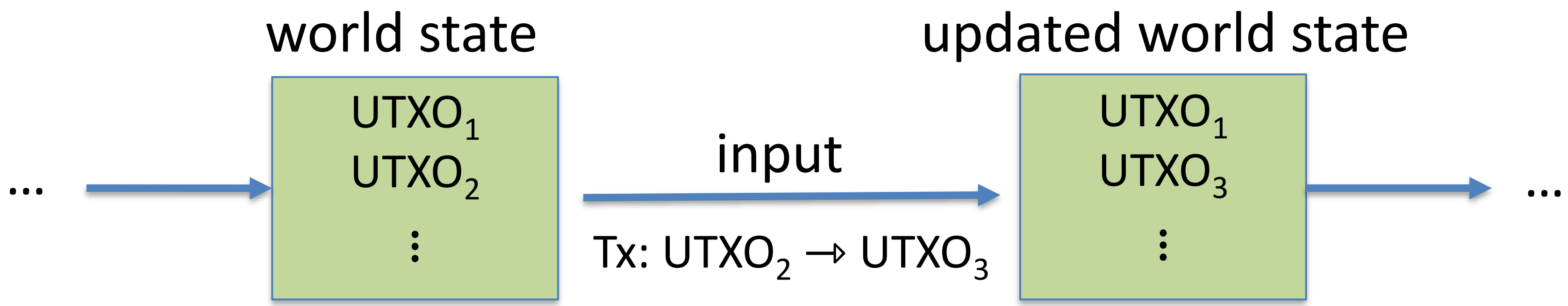
A world of Ethereum Decentralized apps (DAPPs)

- New coins: ERC-20 standard interface
- **DeFi**:   exchanges, lending, stablecoins, derivatives, etc.
- **Insurance**
- **DAOs**: decentralized organizations
- **NFTs**: Managing asset ownership (ERC-721 interface)

⋮

dappradar.com/rankings/protocol/ethereum

Bitcoin as a state transition system



Bitcoin rules: $\boxed{F_{bitcoin} : S \times I \rightarrow S}$

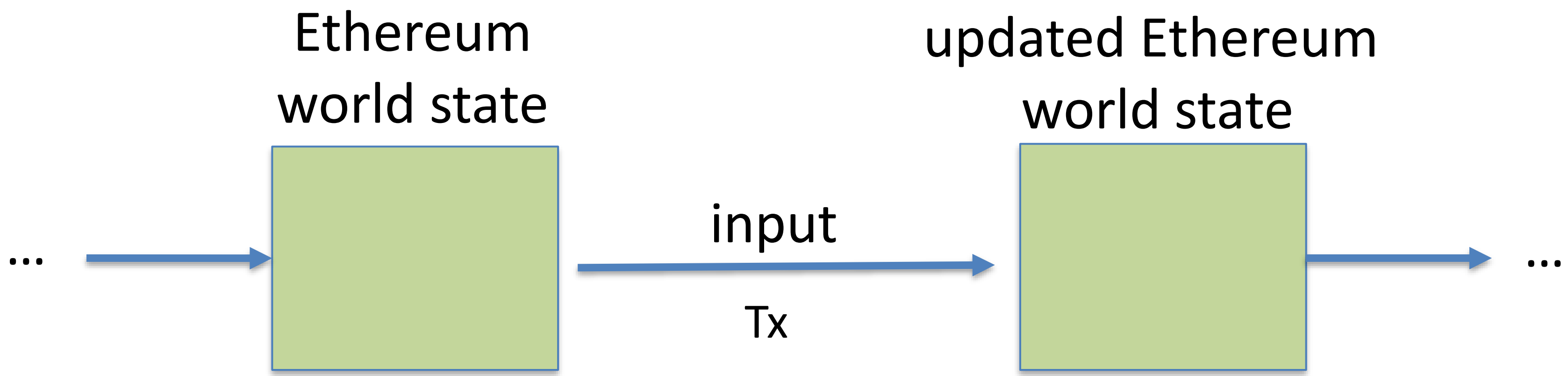
S: set of all possible world states, $s_0 \in S$ genesis state

I: set of all possible inputs

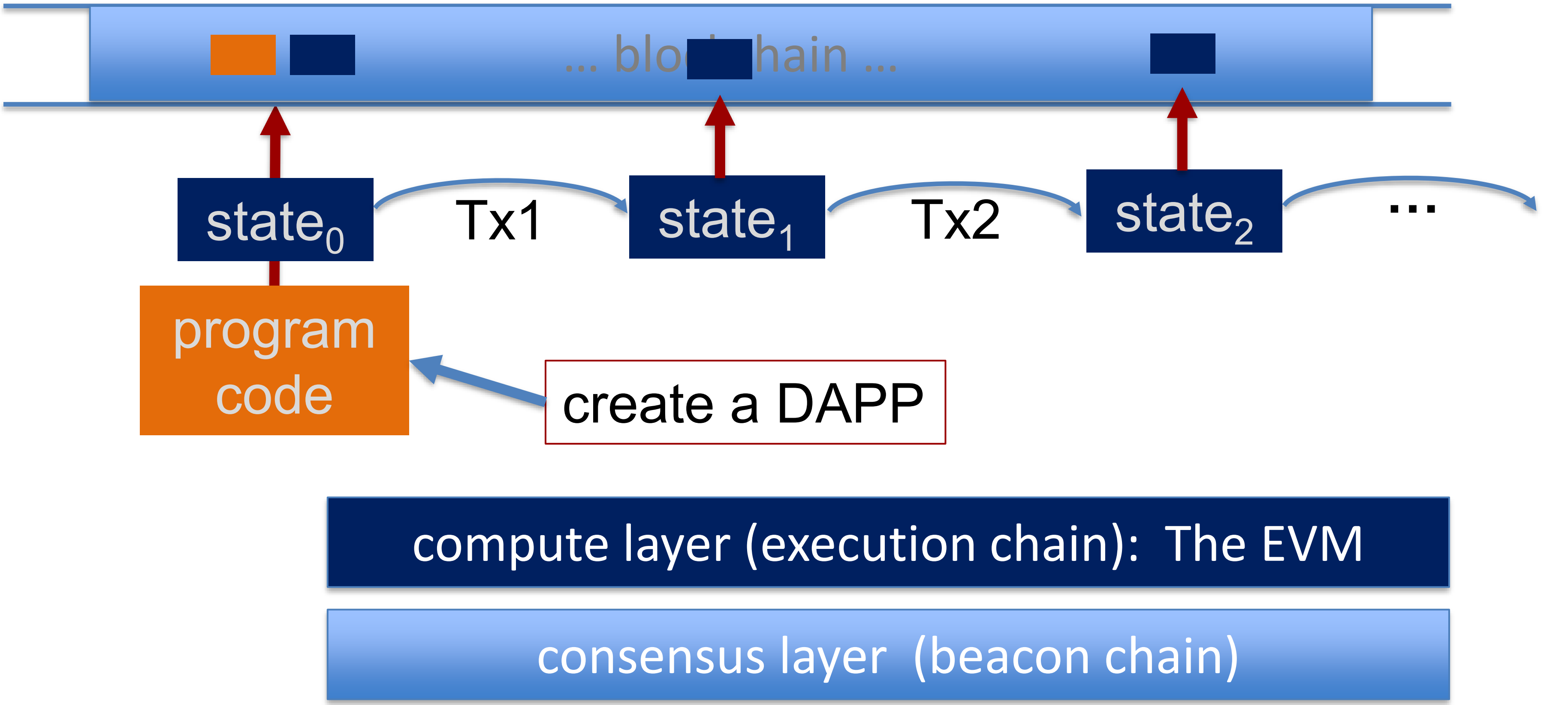
Ethereum as a state transition system

Much richer state transition functions

$\Rightarrow$ one transition executes an entire program



Running a program on a blockchain (DAPP)



The Ethereum system

Proof-of-Stake consensus

Block	Slot	Age	Blobs	Txn	Fee Recipient
23570698	12796465	12 secs ago	6 (67%)	283	Titan Builder
23570697	12796464	24 secs ago	3 (33%)	219	Titan Builder
23570696	12796463	36 secs ago	9 (100%)	285	Fee Recipient: 0xe68...127
23570695	12796462	48 secs ago	4 (44%)	68	beaverbuild
23570694	12796461	1 min ago	3 (33%)	117	Lido: Execution Layer Rew...
23570693	12796460	1 min ago	6 (67%)	206	Titan Builder
23570692	12796459	1 min ago	3 (33%)	278	Titan Builder
23570691	12796458	1 min ago	0 (0%)	187	BuilderNet

source: etherscan.io

One slot every 12 seconds.

(about 200 Tx per block)

One block proposer chosen for each slot (from validator set)

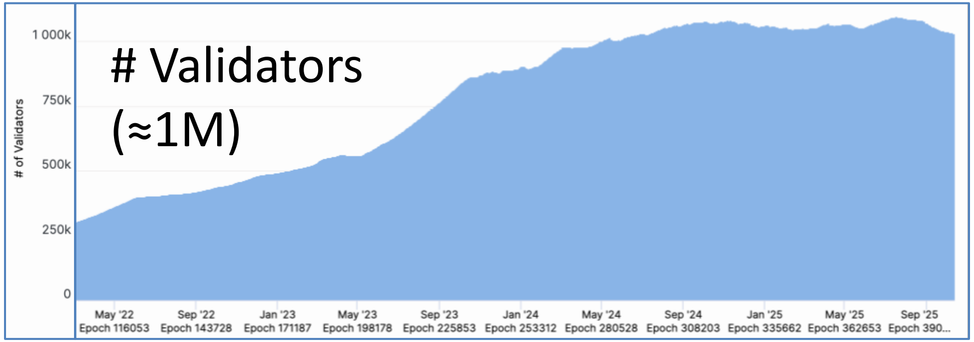
If it sends a valid block, receives Tx fees for block (along with other rewards)

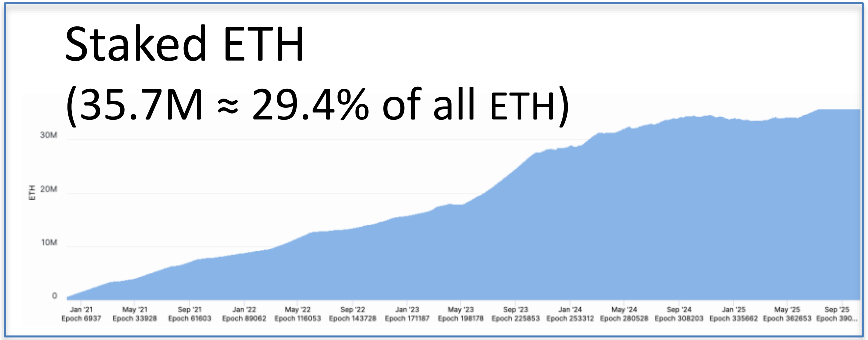
A bit about the beacon chain (Eth2 consensus layer)

To become a validator: stake (lock up) 32 ETH ... or use Lido.

Validators: - sign blocks to express correctness (finalized once enough sigs)

- occasionally act as ***block proposer*** (chosen at random)
- correct behavior ⇒ issued **new ETH** every epoch (32 blocks)
- incorrect behavior ⇒ slashed (lots of details)





The economics of staking

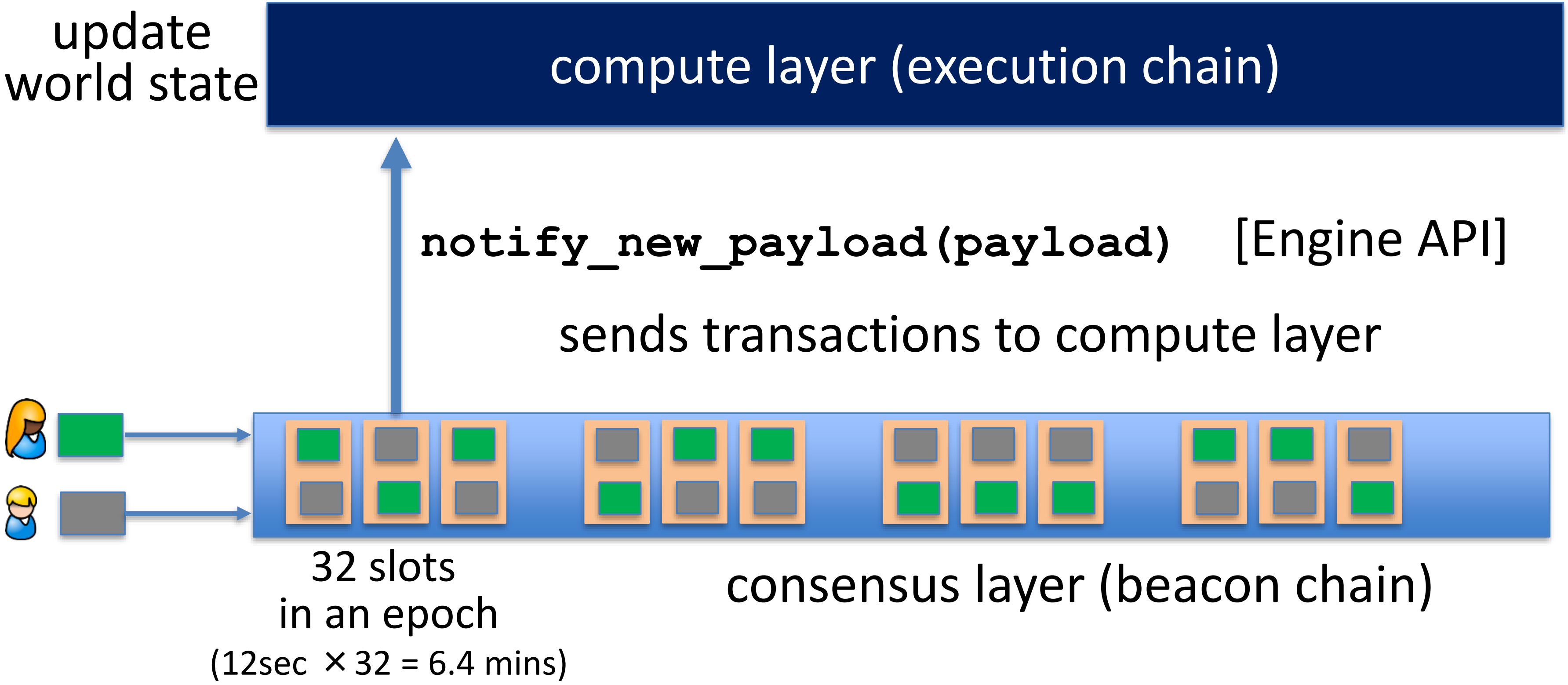
Validator locks up 32 ETH. Oct 2025: 35.7M ETH staked (total)

Annual validator income (an example):

- Sources: issuance, Tx fees (tips), MEV
- Total: 0.93 ETH/year (2.93% on 32 ETH staked)

In practice: staking provider (e.g., Lido) takes a cut of the returns

An Ethereum node



The Ethereum Compute Layer:
The EVM

Ethereum compute layer: the EVM

World state: set of accounts identified by 20-byte address.

Two types of accounts:

(1) externally owned accounts (EOA):

controlled by ECDSA signing key pair (pk,sk).

sk: signing key known only to account owner

Since May 2025: EOA can also be controlled by code

(2) contracts: controlled by code.

code set at account creation time, does not change

Data associated with an account

Account data	Owned (EOA)	Contracts	
address (computed):	H(pk)	H(CreatorAddr, CreatorNonce)	
code:	$\perp$ (or address)	CodeHash	
storage root (state):	$\perp$	StorageRoot	
balance (in Wei):	balance	balance	(1 Wei = 10^{-18} ETH)
nonce:	nonce	nonce	

(#Tx sent) + (#accounts created): anti-replay mechanism

Account state: persistent storage

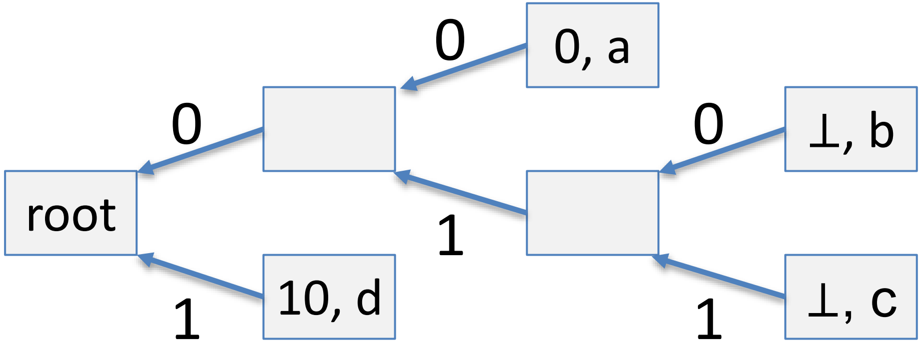
Every contract has an associated **storage array S**[]:

S[0], S[1], … , S[2^{256}-1]: each cell holds 32 bytes, init to 0.

Account storage root: **Merkle Patricia Tree hash** of S[]

- Cannot compute full Merkle tree hash: 2^{256} leaves

S[000] = a
S[010] = b
S[011] = c
S[110] = d



time to compute root hash:
$\leq 2 \times |S|$

|S| = # non-zero cells

State transitions: Tx and messages

<u>Transactions: signed data by initiator</u>

- **To:** 32-byte address of target (0 → create new account)
- **From**, **[Signature]**: initiator address and signature on Tx (if owned)
- **Value**: # Wei being sent with Tx (1 Wei = 10^{-18} ETH)
- Tx fees (EIP 1559)**: gasLimit, maxFee, maxPriorityFee** (later)
- if To = 0: create new contract **code = (init, body)**
- if To ≠ 0: **data** (what function to call & arguments)
- **nonce**: must match current nonce of sender (prevents Tx replay)
- chain_id: ensures Tx can only be submitted to the intended chain

State transitions: Tx and messages

Transaction types:

owned $\rightarrow$ owned: transfer ETH between users

owned $\rightarrow$ contract: call contract with ETH & data

Example (block #10993504)

From		To	msg.value	Tx fee (ETH)
0xa4ec1125ce9428ae5...	→	0x2cebe81fe0dcd220e...	0 Ether	0.00404405
0xba272f30459a119b2...	→	Uniswap V2: Router 2	0.14 Ether	0.00644563
0x4299d864bbda0fe32...	→	Uniswap V2: Router 2	89.839104111882671 Ether	0.00716578
0x4d1317a2a98cfea41...	→	0xc59f33af5f4a7c8647...	14.501 Ether	0.001239
0x29ecaa773f052d14e...	→	CryptoKitties: Core	0 Ether	0.00775543
0x63bb46461696416fa...	→	Uniswap V2: Router 2	0.203036474328481 Ether	0.00766728
0xde70238aef7a35abd...	→	Balancer: ETH/DOUGH...	0 Ether	0.00261582
0x69aca10fe1394d535f...	→	0x837d03aa7fc09b8be...	0 Ether	0.00259936
0xe2f5d180626d29e75...	→	Uniswap V2: Router 2	0 Ether	0.00665809

Messages: virtual Tx initiated by a contract

Same as Tx, but no signature (contract has no signing key)

contract → owned: contract sends funds to user

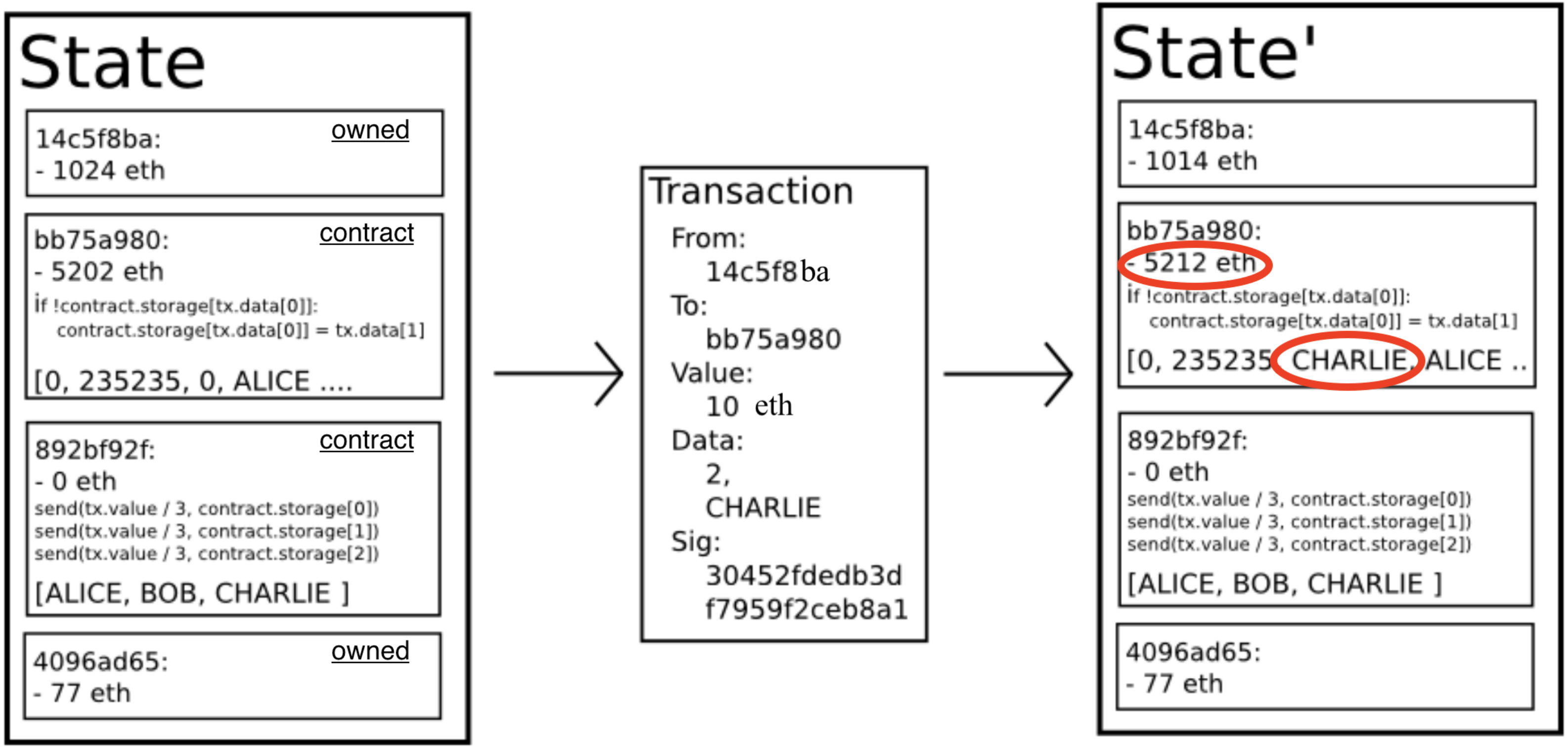
contract → contract: one program calls another (and sends funds)

One Tx from user: can lead to many Tx processed. Composability!

Tx from owned addr → contract → another contract

another contract → different owned

Example Tx



world state (four accounts)

updated world state

An Ethereum Block

Block proposer creates a block of n Tx: (from Txs submitted by users)

- To produce a block do:
  - for i=1,…,n: execute state change of Tx_i sequentially (can change state of >n accounts)
  - record updated world state in block

Other validators re-execute all Tx to verify block $\Rightarrow$ sign block if valid $\Rightarrow$ enough sigs, epoch is finalized.

Block header data (simplified)

(1) consensus data: proposer ID, parent hash, votes, etc.

(2) address of gas beneficiary: where Tx fees will go

(3) world state root: updated world state

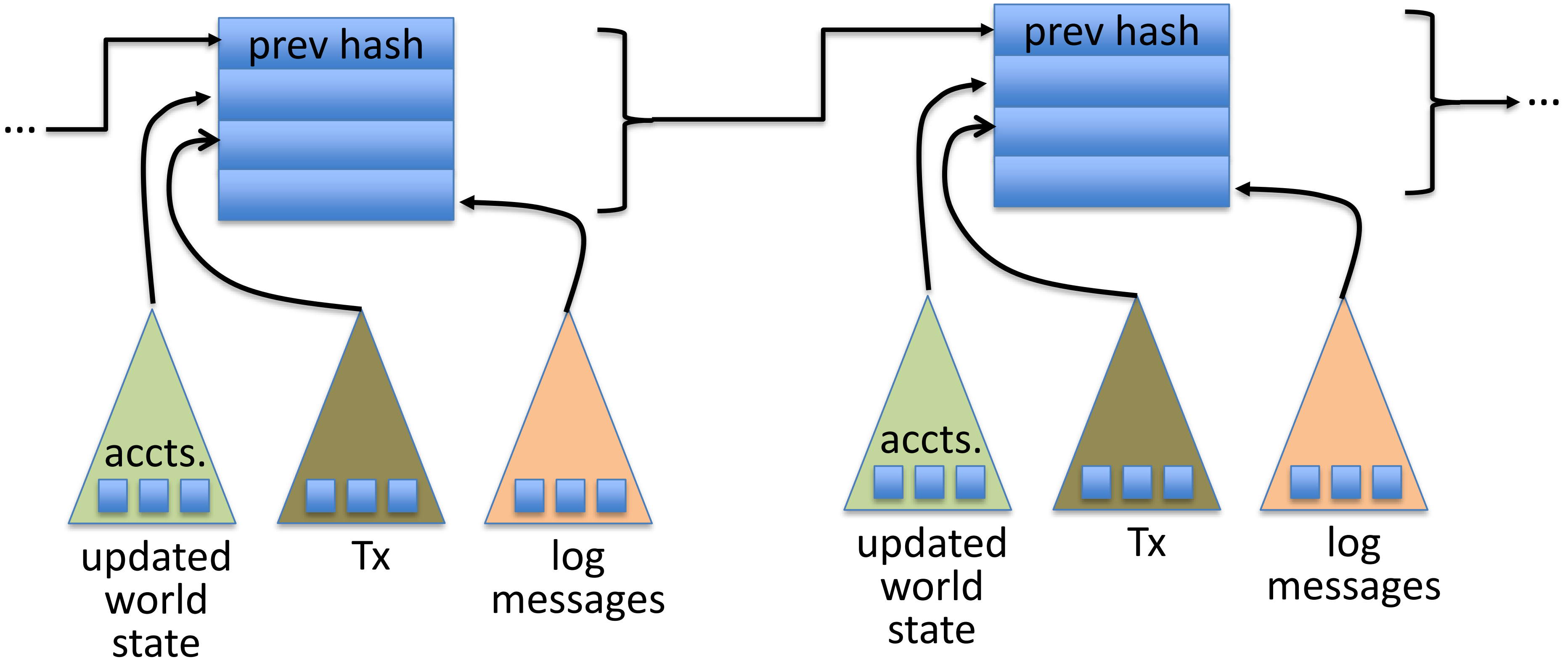
Merkle Patricia Tree hash of <u>all</u> accounts in the system

(4) **Tx root**: Merkle hash of all Tx processed in block

(5) **Tx receipt root**: Merkle hash of log messages generated in block

(6) **Gas Used**: used to adjust the gas price (max gas per block is 45M)

The Ethereum blockchain: abstractly



Amount of memory to run a node

≈1.4 TB

≈1.1 TB

(a year ago)

ETH total blockchain size (archival): 24 TB (Oct. 2025)

An example contract: NameCoin

```
contract nameCoin {          // Solidity code   (next lecture)

    struct nameEntry {
        address owner;       // address of domain owner
        bytes32 value;       // IP address
    }

    // array of all registered domains
    mapping (bytes32 => nameEntry)  data;
```

An example contract: NameCoin

```
function nameNew(bytes32 name) {

    // registration costs is 100 Wei

    if (data[name] == 0   &&   msg.value >= 100) {
        data[name].owner = msg.sender    // record domain owner
        emit Register(msg.sender, name)   // log event
}}
```

Code ensures that no one can take over a registered name

Serious bug in this code!   Front running.   Solved using commitments.

An example contract: NameCoin

```
function nameUpdate(
                bytes32 name, bytes32 newValue, address newOwner) {

  // check if message is from domain owner,
  //             and update cost of 10 Wei is paid

  if (data[name].owner == msg.sender   &&   msg.value >= 10) {

       data[name].value = newValue;         // record new value
       data[name].owner = newOwner;         // record new owner
}}}
```

An example contract: NameCoin

```
function nameLookup(bytes32 name) {
    return data[name];
}

} // end of contract
```

Used by other contracts

Humans do not need this (use etherscan.io)

EVM mechanics: execution environment

Write code in Solidity (or another front-end language)

$\Rightarrow$ compile to EVM bytecode, e.g., using **solc** compiler

(some projects use WASM or BPF bytecode)

$\Rightarrow$ validators use the EVM to execute contract bytecode

in response to a Tx

The EVM

Stack machine (like Bitcoin) but with JUMP

- max stack depth = 1024
- program aborts if stack size exceeded; block proposer keeps gas
- a contract can <u>create</u> or <u>call</u> another contract
  - There are several ways to call another contract
  - Using CALL to call a contract that is different from the caller creates a new volatile execution frame that is deleted on return.

see https://www.evm.codes

The EVM memory types

The EVM has three types of zero initialized memory per contract.

All three are private to the contract that owns them (e.g., nameCoin)

- **Persistent storage** (on blockchain): SLOAD, SSTORE (expensive)
- **Volatile memory** (for a single Tx, one per execution frame):
  MLOAD, MSTORE (very cheap, 3 Gas)
- **Transient memory** (for a single Tx, but behaves like storage):
  TLOAD, TSTORE (cheap, 100 Gas)

- LOG0(data): write data to log

see https://www.evm.codes

Every instruction costs gas, examples:

SSTORE addr (32 bytes), **value** (32 bytes)

- zero → non-zero: 20,000 gas
- non-zero → non-zero: 5,000 gas (for a cold slot)
- non-zero → zero: 15,000 gas refund (example)

Refund is given for reducing size of blockchain state

CREATE : 32,000 + 200 × (code size) gas;

CALL **maxgas**, addr, **value**, args

SELFDESTRUCT addr: kill current contract (5000 gas + 25K gas if addr is empty)

Gas calculation

Why charge gas?

- Tx fees (gas) prevents submitting Tx that runs for many steps.
- During high load: block proposer chooses Tx from mempool that maximize its income.

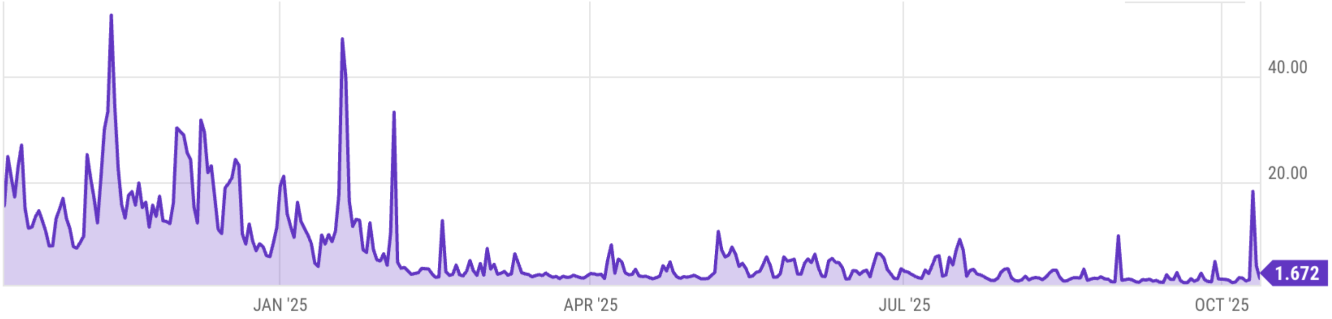
Old EVM: (prior to EIP1559, live on 8/2021)

- Every Tx contains a gasPrice ``bid'' (gas $\rightarrow$ Wei conversion price)
- Producer chooses Tx with highest gasPrice (max sum(gasPrice $\times$ gasLimit))

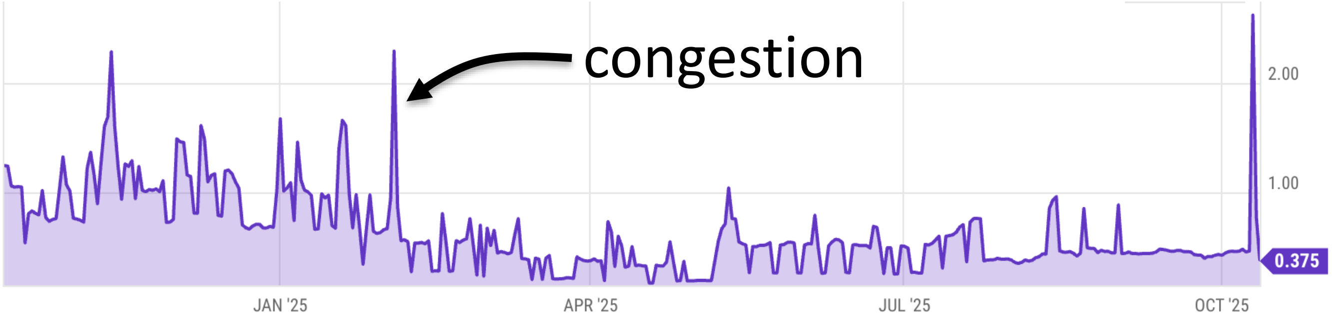
$\Longrightarrow$ not an efficient auction mechanism (first price auction)

Gas prices spike during congestion

GasPrice in Gwei: 1.672 Gwei $= 1.672 \times 10^{-9}$ ETH



Average Tx fee in USD:



Gas calculation: EIP1559 (since 8/2021)

EIP1559 goals (informal):

- users incentivized to bid their true utility for posting Tx,
- block proposer incentivized to not create fake Tx, and
- disincentivize off chain agreements.

[Transaction Fee Mechanism Design, by T. Roughgarden, 2021]

Gas calculation: EIP1559

Every block has a “baseFee”:

the **minimum** gasPrice for all Tx in the block

baseFee is computed from total gas in earlier blocks:

- earlier blocks at gas limit (45M gas) ⟹ base fee goes up 12.5%
- earlier blocks empty ⟹ base fee decreases by 12.5%

interpolate in between

If earlier blocks at “target size” (22.5M gas) ⟹ base fee does not change

Gas calculation

EIP1559 Tx specifies three parameters:

- **gasLimit**: max total gas allowed for Tx
- **maxFee:** maximum allowed gas price (max gas → Wei conversion)
- **maxPriorityFee**: additional "tip" to be paid to block proposer

Computed **gasPrice** bid:

gasPrice ← min(**maxFee**, **baseFee** + **maxPriorityFee)**

Max Tx fee: **gasLimit** × **gasPrice**

Gas calculation (informal)

gasUsed ⇽ gas used by Tx

Send **gasUsed** × (**gasPrice – baseFee**) to block proposer

BURN **gasUsed** × **baseFee** 

⇒ total supply of ETH can decrease

Gas calculation

(1) if **gasPrice** < **baseFee**: abort

(2) If **gasLimit × gasPrice** < msg.sender.balance: abort

(3) deduct **gasLimit × gasPrice** from msg.sender.balance

---

(4) set **Gas** ← **gasLimit**

(5) execute Tx: deduct gas from **Gas** for each instruction

if at end (**Gas** < 0): abort, Tx is invalid (proposer keeps **gasLimit × gasPrice**)

(6) Refund **Gas** × **gasPrice** to msg.sender.balance

---

(7) **gasUsed** ← **gasLimit – Gas**

(7a) BURN **gasUsed × baseFee**

(7b) Send **gasUsed × (gasPrice – baseFee)** to block producer

Why burn ETH ???

Recall: EIP1559 goals (informal)

- users incentivized to bid their true utility for posting Tx,
- block proposer incentivized to not create fake Tx, and
- disincentivize off chain agreements.

Suppose no burn (i.e., baseFee given to block producer):

$\Longrightarrow$ in periods of low Tx volume proposer would try to increase volume by offering to refund the baseFee *off chain* to users.

END OF LECTURE